PHY-275 Fall 2026 Assignment #1

You will receive 1 credit upon complete all the tasks.

Task #1: Fill out the entry poll: https://forms.gle/ow1QBdTzXimuhwye8

Task #2: Refresh your memory with derivatives:

Exercise 1: Calculate the derivatives of each of these functions.

$$f(t) = t^4 + 3t^3 - 12t^2 + t - 6$$
$$g(x) = \sin x - \cos x$$
$$\theta(\alpha) = e^{\alpha} + \alpha \ln \alpha$$
$$x(t) = \sin^2 x - \cos x$$

Exercise 2: The derivative of a derivative is called the second derivative and is written $\frac{d^2 f(t)}{dt^2}$. Take the second derivative of each of the functions listed above.

[i]

Task #3: Refresh your memory with the chain rule:

Exercise 3: Use the chain rule to find the derivatives of each of the following functions.

$$g(t) = \sin(t^2) - \cos(t^2)$$
$$\theta(\alpha) = e^{3\alpha} + 3\alpha \ln(3\alpha)$$
$$x(t) = \sin^2(t^2) - \cos(t^2)$$

For task#2 and task#3, write out each step (when necessary).

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[i] Exercise from Leonard Susskind's *Classical Mechanics: theTheoretical Minimum*

Task #4: Once you work out the Task#2 and Task#3, ask AI (Gemini/Claude/ChatGPT) to check your answer. To proof your efforts, take a screen shot of the prompt and the result.

Task #5: Use google scholar to find a recent paper (2020-2026) from *Nature Communication* that applied the knowledge that is related to **classical mechanics**. Submit the paper name in MLA style citation.